

Machine Learning Notes – Outlier Detection & Unsupervised Learning

Topics Covered

- Outlier Detection
 - Isolation Forest
 - Unsupervised Machine Learning
 - Clustering Algorithms
 - K-Means
 - DBSCAN
 - OPTICS
 - Document Clustering
 - Mini Project
 - Feature Selection in Clustering
 - Mind Map
-

1. Outlier Detection

♦ What are Outliers?

Outliers wo data points hote hain jo baaki dataset se bahut different hote hain.

Example:

Salary
₹25,000
₹30,000
₹28,000
₹45,00,000 ← Outlier

♦ Why Detect Outliers?

- Model accuracy improve hoti hai
 - Noise reduce hota hai
 - Better predictions milti hain
 - Data analysis reliable hota hai
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◆ Causes of Outliers

1. Human Error

Typing mistakes

Example: Age = 300

2. Measurement Error

Sensor/device error

3. Natural Variation

Rare but valid observations

◆ Outlier Detection Methods

Method Type	Examples
Statistical	Z-Score, IQR
Distance-Based	KNN
ML-Based	Isolation Forest

2. Isolation Forest

Isolation Forest ek anomaly detection algorithm hai.

◆ Main Idea

Outliers easily isolate ho jaate hain.

◆ Working of Isolation Forest

Step 1:

Random feature select karo

Step 2:

Random split choose karo

Step 3:

Data recursively split hota hai

Step 4:

Jo point jaldi isolate ho jaye = Outlier

◆ Advantages

- Fast
 - Large datasets support
 - High-dimensional data handle karta hai
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◆ Disadvantages

- Small datasets me less effective
 - Parameter tuning required
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3. Unsupervised Machine Learning

Unsupervised Learning me:

- Data unlabeled hota hai
 - Output available nahi hota
 - Model khud patterns find karta hai
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◆ Types of Unsupervised Learning

Type	Purpose
Clustering	Similar groups banana
Association	Relationships find karna



4. Clustering

Clustering ka goal:

Similar data points ko same group me rakhna.

◆ Applications

- Customer Segmentation
 - Recommendation Systems
 - Image Segmentation
 - Document Grouping
-

◆ Popular Clustering Algorithms

Algorithm	Type
K-Means	Centroid-Based
DBSCAN	Density-Based
OPTICS	Density-Based

⚡ 5. K-Means Clustering

K-Means dataset ko K clusters me divide karta hai.

◆ Working of K-Means

Step 1:

K centroids initialize karo

Step 2:

Nearest centroid assign karo

Step 3:

Centroid update karo

Step 4:

Repeat until convergence

◆ Advantages

- Simple
 - Fast
 - Easy implementation
-

◆ Disadvantages

- K choose karna padta hai
 - Outliers se affect hota hai
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6. DBSCAN

Full Form

Density-Based Spatial Clustering of Applications with Noise

◆ Main Idea

Dense regions ko clusters banata hai aur sparse points ko noise treat karta hai.

◆ Parameters

1. eps

Neighborhood radius

2. MinPts

Minimum points for cluster

◆ Types of Points

Point Type	Meaning
Core Point	Dense region
Border Point	Cluster edge
Noise Point	Outlier

◆ Advantages

- Arbitrary shape clusters detect karta hai
- Noise identify karta hai
- K specify nahi karna padta

◆ Disadvantages

- Parameter tuning difficult
- Different density clusters me problem

7. Document Clustering using DBSCAN

Document clustering me similar documents ko same cluster me rakha jata hai.

◆ Process

Step 1: Text Preprocessing

- Tokenization
- Stopword removal
- Stemming/Lemmatization

Step 2: TF-IDF Vectorization

Documents ko numerical vectors me convert karna.

Step 3: Apply DBSCAN

Similar documents same cluster me group ho jaate hain.

◆ Applications

- News grouping
 - Search engines
 - Topic modeling
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8. OPTICS

Full Form

Ordering Points To Identify Clustering Structure

◆ Main Idea

OPTICS DBSCAN ka advanced version hai.

Ye varying density clusters ko better detect karta hai.

◆ Advantages

- Different density clusters handle karta hai
 - Better cluster structure
 - Noise handling achi hai
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◆ Disadvantages

- Computationally expensive
 - Complex visualization
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9. Mini Project

Feature Selection in Clustering: Comparative Implementation and Performance Analysis

Objective

Different feature selection techniques ka clustering performance par effect analyze karna.

◆ Step 1: Dataset Collection

Example datasets:

- Iris Dataset
 - Customer Dataset
 - Text Dataset
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◆ Step 2: Data Preprocessing

- Missing values handling
 - Scaling
 - Encoding
-

◆ Step 3: Apply Feature Selection

Methods:

- Correlation-Based Selection
 - PCA
 - Variance Threshold
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◆ Step 4: Apply Clustering Algorithms

Algorithms:

- K-Means
- DBSCAN

◆ Step 5: Performance Evaluation

Metrics:

- Silhouette Score
- Davies-Bouldin Index
- Cluster Purity

◆ Comparative Analysis

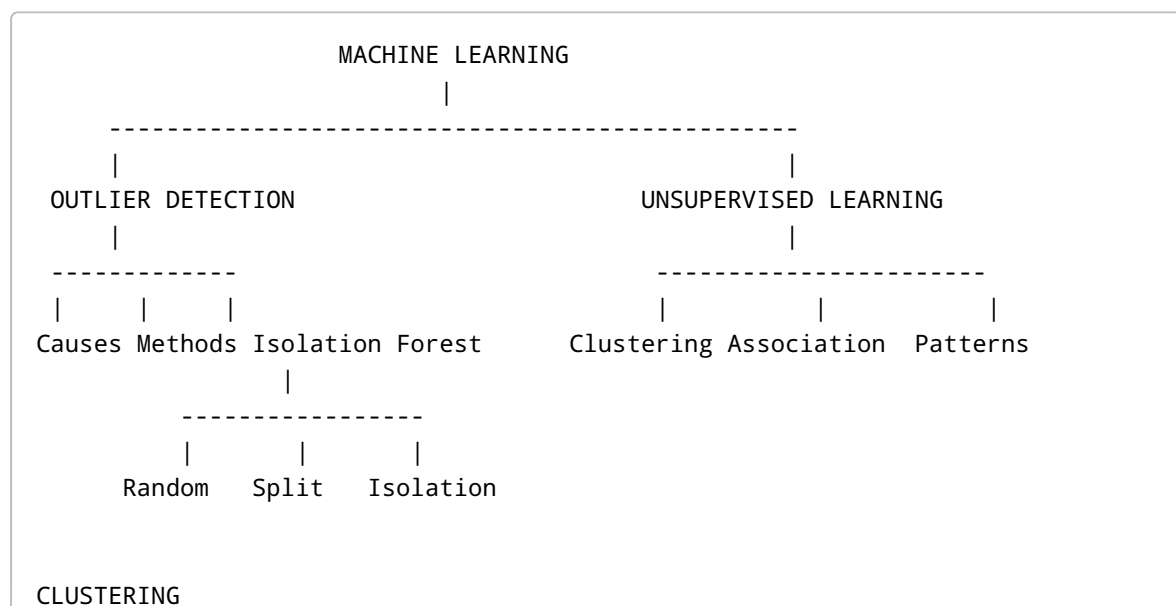
Compare:

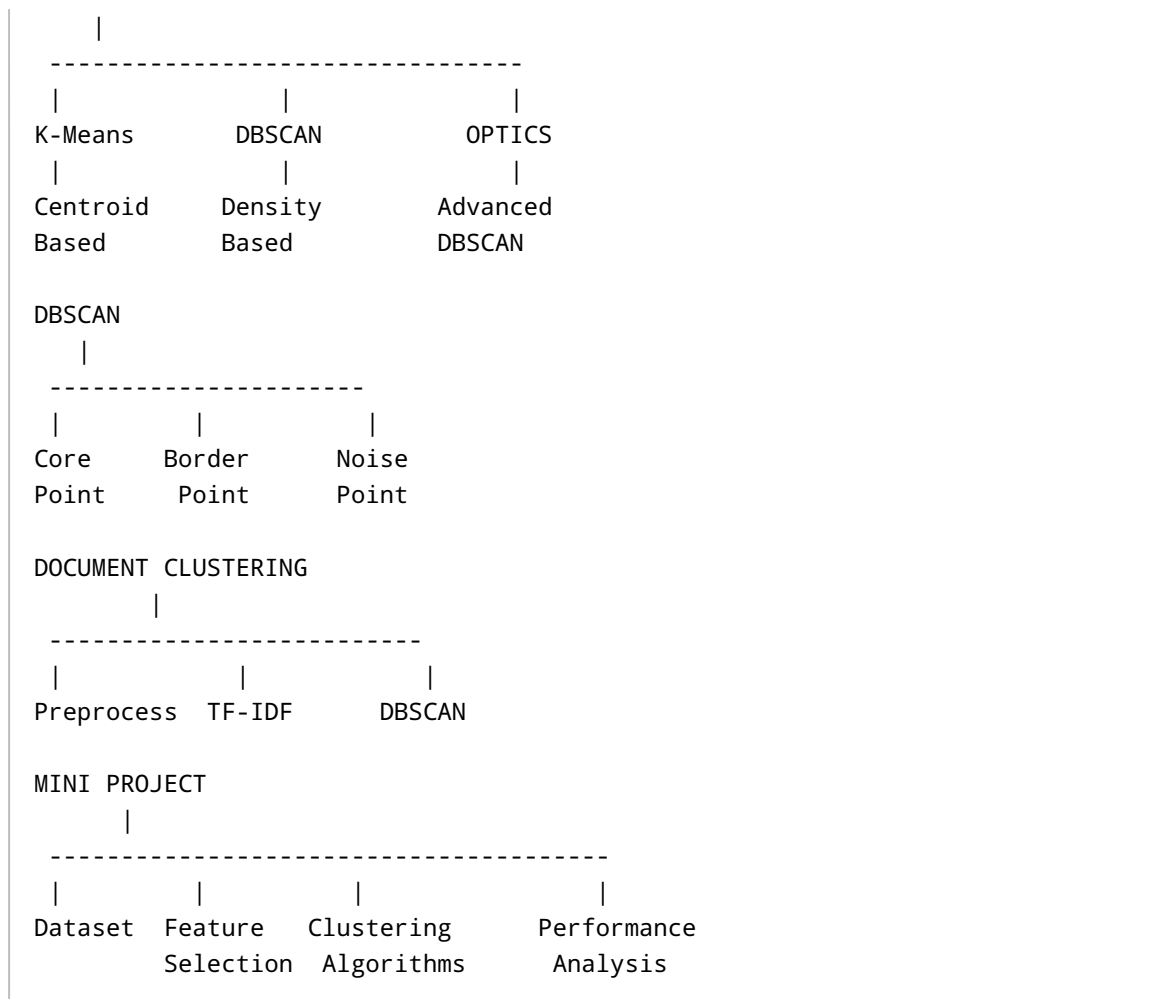
- Accuracy
- Cluster Quality
- Execution Time

Expected Outcome

- Best feature selection method identify hogi
- Clustering quality improve hogi
- Dimensionality reduction ka effect samajh aayega

MIND MAP





1 2 3 4 Numerical Examples of Algorithms

1 K-Means Numerical Example

Given Data Points

Point	Coordinates
A	(1,1)
B	(2,1)
C	(4,3)
D	(5,4)

Assume:

- $K = 2$

Initial Centroids:

- C1 = (1,1)
 - C2 = (5,4)
-

Step 1: Distance Formula

Distance Formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

For Point B(2,1)

Distance from C1:

$$= \sqrt{((2-1)^2 + (1-1)^2)} = \sqrt{1} = 1$$

Distance from C2:

$$= \sqrt{((5-2)^2 + (4-1)^2)} = \sqrt{18} \approx 4.24$$

✅ Since $1 < 4.24$, Point B belongs to Cluster 1.

Step 2: Form Clusters

Cluster 1

- A(1,1)
- B(2,1)

Cluster 2

- C(4,3)
 - D(5,4)
-

Step 3: Update Centroids

Centroid Formula:

$$\text{Centroid} = (\Sigma x / n, \Sigma y / n)$$

New Centroid of Cluster 1

$$C1 = ((1+2)/2, (1+1)/2) = (1.5, 1)$$

New Centroid of Cluster 2

$$C2 = ((4+5)/2, (3+4)/2) = (4.5, 3.5)$$

Repeat until centroids stop changing.

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2 DBSCAN Numerical Example

Given Points

(1,1), (1,2), (2,2), (8,8)

Parameters

- $\epsilon = 2$
- $\text{MinPts} = 2$

Step 1: Neighborhood Check

Distance between (1,1) and (1,2)

$$= \sqrt{((1-1)^2 + (2-1)^2)} = 1$$

✓ Since $1 < \epsilon$, They are neighbors.

Step 2: Cluster Formation

Cluster

- (1,1)
- (1,2)
- (2,2)

Noise Point

- (8,8)

✓ Final Result:

Cluster = {(1,1), (1,2), (2,2)}

Noise = {(8,8)}

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3 OPTICS Numerical Example

Dataset

(1,1), (2,2), (3,3), (10,10)

OPTICS calculates:

- Reachability Distance
 - Core Distance
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Core Distance Example

For point (2,2):

Nearest neighbors:

- (1,1)
- (3,3)

Distance:

$$= \sqrt{((3-2)^2 + (3-2)^2)} = \sqrt{2} \approx 1.41$$

This value helps determine density structure.

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4 Isolation Forest Numerical Example

Dataset

[5, 6, 7, 8, 100]

Here 100 is suspected outlier.

Random Splitting

Suppose split condition:

$$x < 50$$

Then:

Left Side

[5, 6, 7, 8]

Right Side

[100]

✓ 100 isolated immediately.

Hence: 100 = Outlier

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5 PCA Numerical Example

Dataset

Height	Weight
150	50
160	55
170	65

Step 1: Standardization

Mean remove kiya jata hai aur scaling apply hoti hai.

Step 2: Covariance Matrix

$$\text{Cov}(X,Y) = \Sigma[(X-\bar{X})(Y-\bar{Y})] / (n-1)$$

Suppose covariance matrix:

	Height	Weight
Height	1	0.95
Weight	0.95	1

✓ High correlation means PCA dimensions reduce kar sakta hai.

Step 3: Principal Components

PCA converts:

- Height
- Weight

into:

- PC1
- PC2

PC1 contains maximum variance.

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Silhouette Score Numerical Example

Formula

$$S = (b-a) / \max(a,b)$$

Where:

- a = intra-cluster distance
- b = nearest-cluster distance

Given Values

- a = 2
- b = 5

Calculation:

$$S = (5-2) / 5 = 3/5 = 0.6$$

 Higher score means better clustering.

Conclusion

- Outlier Detection important data cleaning step hai.
- Isolation Forest anomalies efficiently detect karta hai.
- Clustering similar data points ko groups me divide karta hai.
- DBSCAN aur OPTICS noise handling me useful hain.
- PCA dimensionality reduce karta hai.

- Feature Selection clustering performance improve karta hai.
- Outlier Detection data cleaning ka important part hai.
- Isolation Forest anomalies efficiently detect karta hai.
- Unsupervised Learning hidden patterns discover karta hai.
- DBSCAN aur OPTICS noise handling aur arbitrary cluster detection me useful hain.
- Feature Selection clustering performance ko significantly improve kar sakta hai.